



**BAKEWISE: Bread Waste Reduction and AI-Driven Knowledge for Efficient
Waste Management, Inventory Analytics, Shelf-Life Prediction, and Production
Enhancement**

In Partial Fulfillment of The
Academic Requirements for the Subject
CAPSTONE RESEARCH 101

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BAKEWISE: Bread Waste Reduction and AI-Driven Knowledge for Efficient Waste Management, Inventory Analytics, Shelf-Life Prediction, and Production Enhancement

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CHAPTER 1: THE PROBLEM AND ITS BACKGROUND

This chapter presents the background of the study, project context, statement of the problem, purpose and description of the proposed system, objectives of the study, scope and limitations, and definition of terms. This chapter establishes the need for the development of BakeWise, an Artificial Intelligence-powered bakery management system designed to reduce bread waste, improve production planning, enhance inventory monitoring, and support data-driven decision-making at Rose Bakeshop – Lasang Branch, Davao City.

The chapter discusses the challenges experienced by bakery businesses in managing products with limited shelf life, maintaining proper inventory levels, and determining accurate production quantities. Through the integration of Artificial Intelligence, Machine Learning, and data analytics, BakeWise aims to provide a technological solution that assists bakery personnel in improving operational efficiency and reducing unnecessary product waste.

1.1 Project Context

1.1.1 Introduction

The advancement of digital technologies has significantly transformed the way organizations manage operations, analyze information, and make business

decisions. Businesses across different industries continue to adopt computerized systems because traditional manual approaches may become inefficient when handling large amounts of operational information and changing market conditions. Digital transformation allows organizations to improve accuracy, increase efficiency, and develop better strategies through the effective use of data.

One of the technologies contributing to modern business improvement is Artificial Intelligence (AI). Artificial Intelligence refers to computer-based technologies capable of analyzing information, identifying patterns, generating predictions, and providing recommendations that assist users in decision-making. Unlike conventional information systems that primarily focus on data storage and retrieval, AI-powered systems are capable of interpreting information and producing meaningful insights based on historical and current data.

According to the United Nations Environment Programme (2024), food waste remains a significant global concern because large quantities of food are lost throughout different stages of production, distribution, and consumption. The issue affects not only environmental sustainability but also the economic performance of businesses involved in food production and service. The report emphasized that improved forecasting, inventory management, and technological integration are important

strategies in reducing unnecessary food waste.

The bakery industry is one of the food sectors that experiences challenges related to waste management because bakery products generally have short shelf lives. Bread products require accurate production planning because producing quantities beyond customer demand may result in unsold products, while insufficient production may lead to missed sales opportunities. Therefore, bakery businesses need effective systems that can analyze demand patterns, monitor product conditions, and support production decisions.

Machine Learning (ML), a branch of Artificial Intelligence, provides opportunities for businesses to improve decision-making by learning from historical data. Machine Learning models can analyze previous sales records, inventory movement, and production information to identify patterns and generate predictions. In bakery operations, these capabilities can support demand forecasting, inventory optimization, and waste reduction strategies (Hübner et al., 2024).

BakeWise applies these technological concepts by developing an AI-powered web-based bakery management system specifically designed for Rose Bakeshop – Lasang Branch, Davao City. The system focuses on improving production planning, monitoring inventory conditions, predicting product shelf life,

analyzing waste patterns, and providing recommendations that support better bakery management decisions.

1.1.2 Background of the Study

Rose Bakeshop is a bakery establishment that provides various baked products, including bread, pastries, cakes, and other food products. For this study, Rose Bakeshop – Lasang Branch in Davao City serves as the official implementation site where BakeWise will be developed, tested, evaluated, and refined based on actual operational requirements.

Like many bakery businesses, Rose Bakeshop – Lasang Branch faces operational challenges related to production planning, inventory monitoring, and management of unsold products. Since bakery products have limited shelf lives, inaccurate estimation of customer demand may lead to excess production, unsold inventory, and unnecessary product waste.

Traditional bakery operations often depend on employee experience and manual estimation when determining production quantities. Although experience remains an important factor in business operations, manual decision-making may not always consider multiple variables such as previous sales trends, customer purchasing behavior, seasonal demand changes, and historical waste patterns.

Another challenge involves inventory monitoring. Without an intelligent monitoring system, bakery personnel may need to manually check product availability and expiration conditions. This process may become difficult when managing multiple products with different shelf-life durations. Delayed identification of near-expiry products may increase the possibility of product loss.

Furthermore, unsold bread products that remain safe for consumption may not always be maximized because businesses may lack systems capable of recommending possible repurposing strategies. Without proper analysis, potentially reusable products may become waste instead of being transformed into alternative bakery products.

The Department of Trade and Industry (2023) emphasized that digital transformation provides opportunities for Micro, Small, and Medium Enterprises (MSMEs) to improve productivity, competitiveness, and operational control. Digital management systems allow businesses to organize information, monitor performance, and make decisions based on accurate data rather than relying solely on manual estimation.

To address these challenges, the researchers propose BakeWise: Bread Waste Reduction and AI-Driven Knowledge for Efficient Waste Management, Inventory Analytics,

Shelf-Life Prediction, and Production Enhancement. The system integrates Artificial Intelligence and Machine Learning technologies to analyze operational data and provide recommendations that improve bakery decision-making.

1.1.3 Adoption Feasibility of Rose Bakeshop

The adoption of BakeWise at Rose Bakeshop – Lasang Branch is feasible because the branch already performs daily operational activities that generate the information required by the system. Sales transactions, production records, inventory monitoring, and waste information provide valuable data sources that can be analyzed by Artificial Intelligence.

The proposed system does not require complete replacement of existing bakery procedures. Instead, BakeWise enhances current workflows by converting operational information into organized digital records. Existing business activities can be improved through automated monitoring, predictive analysis, and recommendation generation.

From a technical perspective, BakeWise is designed as a web-based system that can operate using commonly available devices such as desktop computers, laptops, tablets, and mobile devices. This approach reduces infrastructure requirements and allows gradual

implementation within the bakery environment.

The readiness of bakery personnel is also considered in the adoption process. Employees already possess knowledge regarding product movement, customer demand patterns, production schedules, and daily operations. Through proper training and system orientation, users can learn how to encode information and interpret system-generated recommendations.

The effectiveness of BakeWise depends on several factors, including the availability of accurate operational data, user participation in digital recording procedures, and management support in utilizing system recommendations. The system functions as a decision-support tool where final operational decisions remain under the authority of bakery managers while AI provides analytical assistance.

1.1.4 Necessity of Artificial Intelligence in the Proposed System

Artificial Intelligence is an essential component of BakeWise because bakery operations involve decision-making processes that require prediction, analysis, and continuous improvement. Traditional bakery management approaches commonly rely on employee experience when determining production quantities and handling inventory conditions. Although human expertise remains valuable, manual approaches may not fully

consider complex factors affecting bakery performance.

AI allows businesses to transform historical operational information into meaningful insights. Through Machine Learning algorithms, BakeWise can analyze previous sales records, production information, inventory movement, and waste patterns to identify relationships that support improved decision-making.

The demand forecasting capability of BakeWise allows the system to estimate future product requirements based on historical sales information. Accurate forecasting helps reduce overproduction by allowing bakery personnel to prepare products based on expected customer demand rather than relying only on assumptions.

AI also supports inventory management by analyzing product movement, stock availability, and expiration conditions. Through automated monitoring, the system can identify products that require attention and provide alerts regarding inventory concerns.

Shelf-life prediction is another important AI application in BakeWise. The system evaluates product information, production dates, and shelf-life rules to determine freshness conditions and identify products approaching expiration. This allows bakery personnel to take preventive actions before products become waste.

Additionally, AI supports sustainable food management through product repurposing recommendations. Instead of immediately discarding safe unsold products, BakeWise can suggest possible alternative uses such as bread crumbs, croutons, or bread pudding.

The purpose of AI integration in BakeWise is not to replace human judgment but to improve decision-making accuracy, operational efficiency, and consistency. By combining human expertise with AI-generated insights, bakery businesses can achieve more effective and sustainable operations.

1.2 Statement of the Problem

This study aims to address the operational challenges experienced by Rose Bakeshop – Lasang Branch, Davao City, particularly in production planning, inventory monitoring, shelf-life management, and bread waste reduction.

Specifically, this study seeks to answer the following questions:

1. How can an Artificial Intelligence-powered bakery management system be developed to improve bread waste management and production planning at Rose Bakeshop – Lasang Branch?
2. How can BakeWise assist bakery personnel in forecasting product

demand based on historical sales and production data?

3. How can the system monitor product shelf life and identify products approaching expiration?
4. How can inventory analytics improve monitoring of product availability, movement, and waste patterns?
5. How can AI-generated recommendations assist managers in optimizing production decisions and reducing unnecessary product waste?
6. How can the proposed system support future centralized monitoring and integration of Rose Bakeshop branches under Jathnir Corporation?
7. How acceptable is BakeWise in terms of functionality, usability, reliability, performance efficiency, security, maintainability, and scalability based on software quality evaluation standards?

1.3 Purpose and Description

The purpose of this study is to develop BakeWise, an AI-powered web-based bakery management system designed to reduce bread waste and improve production planning at Rose Bakeshop – Lasang Branch, Davao City.

The system addresses operational problems related to inaccurate demand estimation, excess production, limited inventory monitoring, and difficulties in

managing bakery products with short shelf lives. By integrating Artificial Intelligence, Machine Learning, and data analytics, BakeWise provides bakery personnel with predictive information and recommendations that support better operational decisions.

BakeWise consists of several integrated modules, including demand forecasting, shelf-life prediction, inventory analytics, production optimization, waste analytics, product repurposing recommendation, and management dashboards.

The demand forecasting module analyzes historical sales and production data to estimate future product requirements. The shelf-life prediction module monitors product freshness conditions and identifies products approaching expiration. The inventory analytics module provides information regarding stock movement, availability, and waste trends. The production optimization module recommends appropriate production quantities based on demand forecasts and inventory conditions.

The product repurposing recommendation module provides possible strategies for maximizing safe unsold products instead of immediately disposing of them. Through these features, BakeWise supports sustainable bakery practices and improves operational efficiency.

The system will initially focus on Rose Bakeshop – Lasang Branch. After

successful implementation, validation, and approval of Jathnier Corporation, the system will be integrated into Rose Bakeshop branches within Davao City to support centralized monitoring and company-wide decision-making.

1.4 Objectives of the Study

1.4.1 General Objective

The general objective of this study is to develop **BakeWise, an Artificial Intelligence-powered bakery management system designed to reduce bread waste and improve production planning through demand forecasting, shelf-life prediction, inventory analytics, production optimization, and intelligent product repurposing recommendations at Rose Bakeshop – Lasang Branch, Davao City.**

The proposed system aims to provide bakery personnel and management with a digital decision-support platform that transforms operational data into meaningful insights. Through the integration of Artificial Intelligence and data analytics, BakeWise seeks to improve the accuracy of production decisions, strengthen inventory management practices, minimize unnecessary food waste, and promote sustainable bakery operations.

1.4.2 Specific Objectives

This study specifically aims to develop a demand forecasting feature that analyzes historical sales and production records to predict future bakery product requirements. Through this feature, the system will assist bakery personnel in determining appropriate production quantities based on previous customer purchasing patterns and demand trends.

The study also aims to develop a shelf-life prediction feature that monitors product freshness conditions by analyzing product information, production dates, and shelf-life duration. This feature is intended to provide timely alerts regarding products approaching expiration, allowing bakery personnel to take preventive actions to reduce product loss.

Furthermore, this study aims to create an inventory analytics feature capable of monitoring stock availability, product movement, and inventory trends. Through organized inventory information, bakery managers can identify fast-moving products, slow-moving products, and items that contribute to higher waste levels.

The study also aims to develop a production optimization feature that generates recommendations regarding the quantity of bakery products that should be produced. This feature considers demand forecasts, current inventory conditions, and previous waste patterns to support more accurate production decisions.

Another objective is to develop a product repurposing recommendation feature that provides possible alternative uses for safe unsold bakery products. This feature supports waste reduction by helping maximize the value of products that may otherwise be discarded.

The study also aims to design a dashboard and reporting system that presents sales performance, inventory conditions, waste trends, forecasting results, and AI-generated recommendations in an understandable format for bakery management.

Lastly, this study aims to establish a scalable system structure that can support future integration into other Rose Bakeshop branches within Davao City after successful implementation, evaluation, and approval of Jathnier Corporation.

1.5 Scope and Limitations

1.5.1 Scope of the Study

This study focuses on the design, development, testing, and evaluation of **BakeWise, an AI-powered bakery management system for Rose Bakeshop – Lasang Branch, Davao City**. The system is developed to address operational concerns related to bread waste reduction, demand forecasting, inventory monitoring, shelf-life management, and production planning.

The proposed system covers the collection, storage, processing, and analysis of bakery operational information. The system uses sales records, production data, inventory information, waste records, and product details as inputs for Artificial Intelligence-based analysis.

BakeWise includes a demand forecasting module that analyzes historical sales data to identify product demand patterns and provide future demand predictions. The system also includes shelf-life prediction that monitors product freshness status based on product information and expiration conditions.

The system provides inventory analytics to help monitor available stock, product movement, and waste trends. It also includes production optimization features that generate recommendations regarding appropriate production quantities based on analyzed information.

The waste management component of BakeWise allows bakery personnel to record and analyze unsold, damaged, or expired products. Through waste analytics, the system identifies patterns that may contribute to product loss and provides information that can support waste reduction strategies.

The product repurposing recommendation feature provides possible alternatives for safe unsold bakery products. These

recommendations may include possible conversion into other bakery-related products such as bread crumbs, croutons, and bread pudding.

The system also includes dashboard and reporting features that allow authorized users to monitor bakery performance and generate operational reports. Different user roles are considered, including sales personnel, inventory personnel, production personnel, branch managers, administrators, and future head office users.

The initial implementation of BakeWise will focus only on Rose Bakeshop – Lasang Branch, Davao City. After successful development, testing, validation, and approval from Jathnier Corporation, the system may be integrated into other Rose Bakeshop branches within Davao City to support centralized monitoring and management decision-making.

1.5.2 Limitations of the Study

Although BakeWise provides AI-powered assistance for bakery management, the system has several limitations.

The system does not directly control or automate physical bakery equipment such as ovens, mixers, production machines, or delivery operations. BakeWise functions as a decision-support system that provides

recommendations based on collected information.

The accuracy of AI-generated predictions depends on the quality, completeness, and consistency of available data. Incomplete, incorrect, or outdated records may affect the reliability of forecasting results and recommendations.

The system is initially designed based on the operational requirements of Rose Bakeshop – Lasang Branch. Although the architecture allows future expansion, additional adjustments may be necessary before implementation in other branches due to differences in workflow, product offerings, staffing, and operational procedures.

The system also depends on proper user participation. Bakery personnel must consistently encode accurate sales, production, inventory, and waste information to ensure reliable analysis.

Furthermore, the integration of BakeWise into other Rose Bakeshop branches is subject to the approval, resources, and implementation decisions of Jathnier Corporation. The researchers will only establish the system framework and scalability features required for possible future expansion.

1.6 Definition of Terms

The following terms are defined according to their application in this study.

Artificial Intelligence (AI). Artificial Intelligence refers to computer-based technology that enables systems to analyze information, recognize patterns, generate predictions, and provide recommendations similar to human decision-making processes. In this study, AI is used to support demand forecasting, shelf-life prediction, inventory analytics, and waste reduction strategies.

Bakery Management System. A computerized system designed to support bakery operations by managing information related to sales, production, inventory, product monitoring, and waste management.

BakeWise. The proposed Artificial Intelligence-powered bakery management system developed to reduce bread waste and improve production planning at Rose Bakeshop – Lasang Branch, Davao City.

Bread Waste. Refers to bakery products that are discarded due to overproduction, expiration, damage, unsold inventory, or quality issues.

Centralized Monitoring. A management approach where information from different branches can be collected, analyzed, and monitored through one integrated platform.

Demand Forecasting. The process of predicting future product demand using historical sales data, production records, and identified customer purchasing patterns.

Inventory Analytics. The process of analyzing inventory information to determine stock movement, product availability, waste trends, and management opportunities.

Machine Learning (ML). A branch of Artificial Intelligence that allows computer systems to learn from existing data and improve prediction accuracy through pattern recognition.

Production Optimization. The process of determining appropriate production quantities based on predicted demand, inventory status, and historical operational information.

Product Repurposing. The process of converting safe unsold bakery products into alternative usable products instead of immediately disposing of them.

Shelf-Life Prediction. The process of estimating the remaining freshness period of bakery products using product information, production dates, and shelf-life data.

Waste Analytics. The process of analyzing waste records to identify patterns, causes, and opportunities for reducing product loss.

Decision Support System. A computerized system that assists

managers in making informed decisions by providing analyzed information, reports, and recommendations.

Jathnier Corporation. The organization responsible for Rose Bakeshop operations and the approving body for possible future integration of BakeWise into other Rose Bakeshop branches.

CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES/SYSTEMS

This chapter presents the related literature, studies, and existing systems that serve as the foundation for the development of BakeWise: Bread Waste Reduction and AI-Driven Knowledge for Efficient Waste Management, Inventory Analytics, Shelf-Life Prediction, and Production Enhancement. The discussion focuses on food waste management, Artificial Intelligence, Machine Learning, demand forecasting, shelf-life prediction, inventory optimization, and digital transformation in food-related businesses.

The reviewed studies provide evidence that Artificial Intelligence and data-driven technologies can improve operational efficiency by allowing organizations to analyze historical information, predict future conditions, and support decision-making. The chapter also identifies existing limitations in current systems and establishes the research gap addressed by BakeWise.

2.1 Review of Related Literature

2.1.1 Philippine Food Waste Situation

Food waste continues to be a significant concern in the Philippines because of its economic, environmental, and social impacts. Food losses occur across different stages of the food supply chain, including production, distribution, retail, and consumption. The Philippine Statistics Authority (2023) reported that inefficient food management practices contribute to resource losses and highlight the need for improved systems that support proper planning and utilization of food resources.

Food businesses, including bakeries, are particularly affected because many products have limited shelf lives. Bread products require accurate production planning because excessive production may result in unsold products while insufficient production may affect customer satisfaction and sales opportunities. Without proper forecasting and monitoring systems, businesses may experience difficulties balancing supply and demand.

The Food and Agriculture Organization (FAO, 2019) emphasized that reducing food loss and waste requires improvements in planning, storage, monitoring, and decision-making processes. Technology-driven solutions provide opportunities for businesses to better understand demand patterns and optimize resource utilization.

For bakery businesses, digital systems can assist in identifying products with high waste rates, monitoring inventory conditions, and improving production decisions. These concepts support the development of BakeWise because the proposed system focuses on reducing bread waste through Artificial Intelligence-based forecasting, inventory analytics, and waste monitoring.

2.1.2 Digital Transformation of Philippine MSMEs

Digital transformation has become an important strategy for improving the competitiveness of Micro, Small, and Medium Enterprises (MSMEs). According to the Department of Trade and Industry (2023), adopting digital technologies enables businesses to improve operational efficiency, strengthen information management, and make more informed decisions.

Many MSMEs still rely on manual processes for recording transactions, monitoring inventory, and managing daily operations. While manual approaches may work for small-scale activities, they may become inefficient when businesses handle increasing amounts of information and changing customer demands.

Digital management systems provide organizations with structured information that can be analyzed for business improvement. Through automation and data analytics, businesses can identify

trends, monitor performance, and improve resource allocation.

In the bakery industry, digital transformation can improve areas such as production scheduling, inventory management, and waste reduction. BakeWise applies this concept by converting traditional bakery records into digital information that can be analyzed through Artificial Intelligence to generate useful recommendations.

2.1.3 Technology Adoption in Philippine Businesses

The adoption of advanced technologies among Philippine businesses provides opportunities for improving productivity and operational performance. However, challenges such as limited technical knowledge, financial resources, and infrastructure readiness may affect the implementation of digital solutions (Department of Science and Technology, 2022).

Despite these challenges, accessible technologies such as cloud-based systems, data analytics, and Artificial Intelligence applications allow businesses to benefit from digital transformation without requiring highly complex infrastructure.

The successful adoption of technology depends on system usability, employee readiness, management support, and alignment between technology and business needs. Systems designed according to actual operational

requirements are more likely to be accepted and effectively utilized.

BakeWise considers these factors by developing a user-friendly web-based system designed specifically for bakery operations. The system focuses on practical AI implementation that supports existing workflows rather than replacing human decision-making.

2.1.4 Bakery Waste Management Challenges in Davao City

Bakery businesses in urban areas such as Davao City face operational challenges related to production planning, inventory control, and management of unsold products. Bread products require careful monitoring because their quality decreases over time, making accurate forecasting essential.

Traditional bakery operations often depend on employee experience when estimating daily production requirements. Although experienced employees can provide valuable insights, manual estimation may not always consider changing customer behavior, seasonal demand variations, and historical sales patterns.

Waste management becomes challenging when businesses lack tools that can identify the causes of recurring product loss. Without proper analysis, businesses may only record waste after it occurs instead of preventing it through predictive decision-making.

The development of BakeWise addresses these challenges by providing a system capable of analyzing sales patterns, predicting demand, monitoring product freshness, and recommending strategies to reduce unnecessary bread waste.

2.1.5 Artificial Intelligence in the Food Service Industry

Artificial Intelligence has become increasingly significant in the food service industry because of its ability to improve efficiency, forecasting accuracy, and operational decision-making. AI technologies allow organizations to analyze large amounts of data and generate recommendations based on identified patterns.

According to Dwivedi et al. (2021), Artificial Intelligence applications support organizations by improving prediction capabilities, automating repetitive processes, and enhancing decision support. In food-related industries, AI can be applied in demand forecasting, inventory management, customer analysis, and waste reduction.

AI-based systems provide advantages because they can process information faster and identify patterns that may not be immediately visible through manual analysis. These capabilities are particularly valuable in industries involving perishable products where timing and accuracy are essential.

BakeWise applies Artificial Intelligence through demand forecasting, shelf-life prediction, inventory analytics, and waste analysis. These functions demonstrate how AI can support sustainable and efficient bakery operations.

2.2 Foreign Literature

2.2.1 Global Food Waste Crisis

Food waste is recognized as a global challenge affecting sustainability, economic efficiency, and environmental protection. The United Nations Environment Programme (2024) reported that significant amounts of food continue to be wasted worldwide due to inefficient planning, poor inventory management, and limited monitoring systems.

The report emphasized the importance of improving food management practices through better measurement, prevention strategies, and technological solutions. Businesses that handle perishable products require systems capable of predicting demand and monitoring product conditions to minimize unnecessary losses.

The global concern regarding food waste provides justification for developing technology-based solutions such as BakeWise. By applying AI-driven analytics, the proposed system supports efforts to reduce bakery waste at the operational level.

2.2.2 Machine Learning for Bakery Demand Forecasting

Demand forecasting plays an important role in bakery operations because production decisions depend heavily on customer demand patterns. Hübner et al. (2024) examined the use of Machine Learning approaches in forecasting bakery product demand and found that predictive models can improve production planning by analyzing historical sales information.

Machine Learning models can identify patterns from previous sales records and generate predictions that assist businesses in determining appropriate production quantities. This reduces the risk of overproduction and helps maintain product availability.

The findings of Hübner et al. (2024) support the demand forecasting feature of BakeWise, where historical bakery data are analyzed to provide production recommendations.

2.2.3 Machine Learning-Based Shelf-Life Prediction

Shelf-life prediction is an important application of Artificial Intelligence in food management because it helps organizations determine product freshness and prevent unnecessary disposal.

Research by Wunderlich et al. (2023) demonstrated that Machine Learning models can assist in predicting food shelf life by analyzing product-related

information and environmental factors. Predictive models provide businesses with early warnings regarding product conditions.

This concept supports the shelf-life prediction module of BakeWise, which aims to identify bakery products approaching expiration and provide timely recommendations for action.

2.2.4 Digital Inventory Optimization Systems

Inventory management is a critical factor in businesses handling perishable products. Melesse et al. (2022) emphasized that digital inventory systems improve visibility of product movement, stock conditions, and resource utilization.

Traditional inventory management may create challenges when businesses need to monitor large amounts of information manually. Digital systems provide better tracking capabilities and allow organizations to make decisions based on accurate data.

BakeWise incorporates inventory analytics to improve stock monitoring and identify conditions that contribute to waste.

2.2.5 Artificial Intelligence-Based Food Waste Monitoring

Artificial Intelligence-based waste monitoring systems provide opportunities for organizations to

understand waste patterns and develop prevention strategies. Rokhva et al. (2025) highlighted that AI technologies can support food waste analysis by identifying trends and providing information that assists sustainability efforts.

Instead of only recording waste after it occurs, intelligent systems allow businesses to analyze causes and develop preventive actions.

This supports the waste analytics component of BakeWise, which aims to identify product waste patterns and provide recommendations for reducing unnecessary losses.

2.3 Local Studies and Existing Systems

2.3.1 Bakery Demand Forecasting System

Demand forecasting systems have become increasingly important in food businesses because accurate prediction of customer demand directly affects production efficiency and inventory management. In bakery operations, inaccurate estimation of customer demand may result in excessive production, unsold products, and unnecessary waste.

Hakim et al. (2024) developed a machine learning-based bakery demand forecasting system that utilized historical sales data to predict future product requirements. The study demonstrated

that predictive models can assist bakery businesses in improving production decisions by identifying patterns from previous sales information. The researchers emphasized that data-driven forecasting provides better accuracy compared to relying only on manual estimation.

The study is relevant to BakeWise because demand forecasting is one of the core features of the proposed system. Similar to the previous study, BakeWise analyzes historical bakery data to generate future demand predictions. However, BakeWise extends beyond forecasting by integrating additional functions such as inventory analytics, shelf-life prediction, waste monitoring, and product repurposing recommendations.

2.3.2 Food Waste Monitoring System

Food waste monitoring systems provide organizations with tools to measure, analyze, and reduce unnecessary food disposal. These systems are important because identifying waste patterns allows businesses to develop strategies that prevent recurring losses.

Caña et al. (2023) developed a food waste monitoring system that applied intelligent data analysis to identify waste patterns and provide information for waste reduction efforts. The study emphasized that technology-based monitoring systems allow organizations to understand the causes of food waste

rather than simply recording disposal activities.

The study supports the development of BakeWise because waste analytics is an essential component of the proposed system. BakeWise incorporates waste recording and analysis features to determine which products contribute most to waste and provide recommendations that help bakery managers improve production and inventory decisions.

2.3.3 Cafeteria Food Waste Analysis

Food waste studies in institutional food environments demonstrate that poor planning and inaccurate estimation contribute significantly to unnecessary food loss. Favis et al. (2022) investigated food waste practices in cafeteria operations and found that production planning, consumer demand estimation, and inventory management influence the amount of food discarded.

The study highlighted the importance of using systematic approaches to monitor food consumption patterns and improve preparation decisions. Data analysis allows organizations to identify waste causes and implement preventive strategies.

Although the study focused on cafeteria operations, its findings are applicable to bakery businesses because both environments involve preparing food based on expected customer demand. BakeWise applies similar principles by

analyzing sales, production, and waste information to support better production planning.

2.3.4 IoT-Based Food Inventory System

Inventory management is a critical aspect of food businesses because products have limited usability periods and require proper monitoring. Technology-based inventory systems provide improved visibility of stock conditions and allow businesses to respond quickly to inventory concerns.

Ortiz et al. (2023) developed an Internet of Things (IoT)-based food inventory monitoring system that focused on improving stock tracking and reducing food loss. The system demonstrated that digital monitoring technologies can improve inventory accuracy and support efficient resource management.

The study supports the inventory analytics feature of BakeWise because both systems focus on improving product monitoring. However, BakeWise incorporates Artificial Intelligence capabilities that extend beyond inventory tracking by providing demand forecasts, shelf-life analysis, and production recommendations.

2.3.5 Bakery Forecasting Model Using CRISP-DM

The Cross Industry Standard Process for Data Mining (CRISP-DM) is a structured methodology used in

developing data-driven systems. Hakim et al. (2024) applied the CRISP-DM approach in creating a bakery forecasting model by following systematic stages involving data understanding, preparation, modeling, evaluation, and implementation.

The study emphasized that successful AI-based systems require organized data collection and proper analysis procedures. The methodology ensures that predictive models are developed using reliable information and evaluated before actual implementation.

This study supports the development approach of BakeWise because the proposed system also relies on historical bakery information for AI analysis. Proper data preparation and evaluation are necessary to ensure that demand forecasting and recommendation features provide reliable results.

2.4 Foreign Studies and Existing Systems

2.4.1 Machine Learning Bakery Forecasting System

Machine Learning has been increasingly applied in bakery operations to improve demand prediction and production planning. Hübner et al. (2024) examined the application of machine learning models in bakery forecasting and found that predictive algorithms can improve operational decisions by identifying

demand patterns from historical information.

The study showed that forecasting accuracy helps businesses reduce production errors and minimize unsold products. By understanding customer demand trends, bakeries can produce appropriate quantities while maintaining product availability.

The findings support the demand forecasting capability of BakeWise. However, the proposed system expands this concept by integrating forecasting with other bakery management functions, including inventory analytics, shelf-life monitoring, and waste reduction strategies.

2.4.2 Food Spoilage Prediction System

Food spoilage prediction systems use Artificial Intelligence and Machine Learning techniques to estimate product conditions and prevent unnecessary disposal. Mamidala (2023) developed a machine learning-based food spoilage prediction system that analyzed product information and environmental factors to determine possible spoilage conditions.

The study demonstrated that predictive technologies can help businesses identify food products requiring immediate attention. Early detection allows organizations to prioritize consumption, adjust inventory strategies, or apply alternative utilization methods.

This study is related to BakeWise because the proposed shelf-life prediction module applies similar principles. BakeWise uses product information, production dates, and shelf-life data to monitor freshness conditions and provide alerts for products approaching expiration.

2.4.3 Deep Learning for Food Demand Forecasting

Deep learning techniques provide advanced capabilities in analyzing complex patterns within large datasets. Ghosh (2025) explored the application of deep learning models for food demand forecasting and demonstrated that advanced predictive approaches can improve demand estimation by analyzing historical consumption patterns.

The study emphasized that accurate forecasting supports better resource allocation, production planning, and inventory management.

The findings support the AI forecasting component of BakeWise because bakery operations require accurate demand estimation to prevent overproduction and reduce waste. By applying predictive analysis, BakeWise aims to assist bakery managers in making informed production decisions.

2.4.4 AI-Based Inventory Optimization Framework

Inventory optimization systems using Artificial Intelligence help organizations maintain appropriate stock levels while minimizing unnecessary losses. Fatima and Salam (2026) developed an AI-based inventory optimization framework that analyzed demand patterns, inventory conditions, and operational information to improve stock management.

The study found that AI-driven inventory systems provide better decision support compared to traditional manual monitoring methods because they can identify patterns and recommend appropriate actions.

This study supports the inventory optimization goals of BakeWise. The proposed system applies similar principles by analyzing product movement, inventory status, and waste patterns to assist bakery managers in improving stock management.

2.4.5 Computer Vision Food Waste System

Computer vision technology has also been applied in food waste management by allowing systems to identify and analyze discarded food products. Rokhva et al. (2025) developed an Artificial Intelligence-based food waste monitoring system using intelligent recognition techniques to understand waste behavior.

The study demonstrated that AI technologies can provide valuable information regarding waste sources and patterns. Such information allows organizations to create more effective waste reduction strategies.

Although BakeWise does not primarily use computer vision technology, the study supports the importance of Artificial Intelligence in food waste analysis. BakeWise applies AI through forecasting, analytics, and recommendation systems to reduce bakery waste through preventive decision-making.

2.5 Synthesis and Research Gap

The reviewed literature and studies demonstrate that Artificial Intelligence, Machine Learning, predictive analytics, and digital management systems have significant potential in improving food industry operations. Existing research confirms that technology-based solutions can improve demand forecasting, inventory management, shelf-life monitoring, and waste reduction strategies.

Studies related to bakery forecasting, such as those conducted by Hübner et al. (2024) and Hakim et al. (2024), established that Machine Learning models can improve production planning by analyzing historical sales data. These studies demonstrate that predictive technologies can reduce production errors and help businesses align output with customer demand.

Research related to shelf-life prediction, including Wunderlich et al. (2023) and Mamidala (2023), showed that Artificial Intelligence can assist businesses in identifying product conditions and preventing unnecessary spoilage. These studies emphasize the importance of early detection and predictive monitoring for businesses handling perishable products.

Inventory optimization studies by Melesse et al. (2022) and Fatima and Salam (2026) further demonstrated that digital inventory systems improve stock visibility and operational efficiency. These studies highlight the importance of accurate information management in reducing resource losses.

Despite these advancements, existing systems commonly focus on only one specific operational concern. Some systems concentrate on demand forecasting, while others focus on inventory monitoring, spoilage prediction, or waste analysis. Few studies integrate these functions into one complete bakery management system.

Furthermore, many existing solutions are designed for large-scale industries and may require advanced infrastructure, making them less suitable for small and medium-sized bakery businesses. There is a need for an accessible, integrated, and practical system that combines Artificial Intelligence capabilities while

considering the operational realities of bakery establishments.

BakeWise addresses this research gap by developing an integrated AI-powered bakery management system that combines demand forecasting, shelf-life prediction, inventory analytics, production optimization, waste analysis, and product repurposing recommendations into one platform.

Unlike existing systems that focus on isolated functions, BakeWise provides a complete decision-support solution designed specifically for bakery operations. The system will initially be implemented at Rose Bakeshop – Lasang Branch, Davao City, where it will be tested and refined based on actual operational requirements. After successful implementation, validation, and approval from Jathnier Corporation, the system may be expanded to other Rose Bakeshop branches within Davao City.

Through this integrated approach, BakeWise contributes to the advancement of sustainable food management by demonstrating how Artificial Intelligence can support waste reduction, improve operational efficiency, and enhance decision-making in the bakery industry.

CHAPTER 3: TECHNICAL BACKGROUND

This chapter presents the technical foundation of BakeWise: Bread Waste

Reduction and AI-Driven Knowledge for Efficient Waste Management, Inventory Analytics, Shelf-Life Prediction, and Production Enhancement. It discusses the development tools, technical description, system requirements, system features, and system architecture of the proposed Artificial Intelligence-powered bakery management system.

The technical background provides an explanation of the technologies, frameworks, databases, and Artificial Intelligence components used in developing BakeWise. These technologies were selected based on their ability to support a reliable, scalable, and user-friendly system that can assist bakery operations.

BakeWise is designed as a web-based intelligent management system that integrates Artificial Intelligence, Machine Learning, database management, and data analytics. Unlike traditional record-keeping systems that only store operational information, BakeWise analyzes collected data and generates recommendations that support production planning, inventory management, and waste reduction.

The system is initially developed and implemented for Rose Bakeshop – Lasang Branch, Davao City. The architecture of the system is designed to allow future expansion and integration into other Rose Bakeshop branches within Davao City after successful

implementation, evaluation, and approval of Jathnier Corporation.

3.1 Development Tools

The development of BakeWise requires various software tools, programming technologies, database platforms, and Artificial Intelligence resources. These tools support the development of the system interface, data management, AI processing, analytics, and future deployment.

The selected technologies are based on their capability to provide efficient development, maintainability, security, and scalability.

Table 1. Development Tools Used in BakeWise

Development Tool/Technology	Purpose in BakeWise
JavaScript	Used to develop interactive and responsive system interfaces. It enables dynamic dashboards, forms, user interactions, and real-time display of bakery information.
Python	Used for Artificial Intelligence and Machine Learning development. It supports data

	processing, forecasting models, predictive analysis, and recommendation generation.
HTML5	Provides the structural framework of the BakeWise web application, including pages, forms, dashboards, and system layouts.
CSS3	Used to design the visual appearance of the system and create responsive layouts suitable for desktop and mobile devices.
React.js / Next.js	Used for front-end development by creating reusable interface components, improving system performance, and supporting responsive web design.
Node.js	Handles server-side operations, application processing, and communication between the user interface and database.

Python AI Libraries (Pandas, NumPy, Scikit-learn, TensorFlow)	Supports Artificial Intelligence functions such as data preparation, machine learning model development, forecasting, and predictive analytics.
MySQL / PostgreSQL	Serves as the database management system for storing sales records, production information, inventory data, waste records, product information, and user accounts.
Chart.js / Recharts	Used to create visual representations such as charts, graphs, and dashboards for easier interpretation of bakery performance data.
Visual Studio Code	Used as the development environment for writing, testing, debugging, and managing system source codes.
Git and GitHub	Used for version control, source code organization, and tracking system development progress.

Cloud Hosting Platform	Provides online accessibility and supports future centralized monitoring among Rose Bakeshop branches.
Web Browser	Allows users to access BakeWise through commonly available browsers without installing additional applications.

3.1.1 Programming Language

Programming languages serve as the foundation of BakeWise because they allow developers to create system functions, process information, and establish communication between different system components.

JavaScript is used for the development of the user interface because it supports interactive and responsive web applications. Through JavaScript, BakeWise can provide users with dynamic dashboards, data entry forms, notifications, and interactive reports. This allows different users, such as sales personnel, inventory staff, production staff, and managers, to efficiently interact with the system.

Python is used for Artificial Intelligence and Machine Learning development because of its strong capabilities in data analysis and predictive modeling.

Python provides various libraries that allow the system to process historical bakery information, develop forecasting models, and generate recommendations.

The combination of JavaScript and Python allows BakeWise to achieve both an accessible user interface and intelligent analytical capabilities.

3.1.2 Web Development Framework

The web development framework provides the structure needed to build and organize the BakeWise application. It manages communication between the user interface, system logic, database, and Artificial Intelligence modules.

A modern web framework allows developers to create organized components, improve system performance, and maintain the application efficiently.

The framework supports important system functions such as user authentication, data processing, dashboard generation, report creation, and integration of AI services.

The use of a web-based framework also improves accessibility because users can access BakeWise through different devices including computers, tablets, and mobile phones.

3.1.3 Database Management System

The Database Management System serves as the storage foundation of BakeWise. It organizes and manages operational information required for bakery monitoring and Artificial Intelligence analysis.

The database stores information related to sales, production, inventory, waste, products, users, forecasts, and generated reports.

Table 2. Database Components of BakeWise

Database Component	Purpose in BakeWise
User Database	Stores user accounts, passwords, roles, and permissions to support secure login and access control.
Product Database	Stores product information including product name, category, price, and shelf-life details.
Sales Database	Stores transaction history and sales patterns used for demand forecasting.
Production Database	Stores production schedules, planned quantities, and actual production records.

Inventory Database	Stores stock levels, product movement, and availability information.
Waste Database	Stores discarded product information, waste quantity, and waste causes for analysis.
Forecast Database	Stores AI-generated predictions and recommendations for management review.
Report Database	Stores generated reports and analytical results for future reference.

The database structure ensures that information collected from daily bakery operations can be organized and analyzed effectively.

3.1.4 Artificial Intelligence and Machine Learning Components

Artificial Intelligence serves as the main technology that allows BakeWise to provide predictive and analytical capabilities.

The system applies Machine Learning techniques to identify patterns from historical bakery information and generate useful recommendations.

Table 3. Artificial Intelligence Components of BakeWise

AI Component	Purpose in BakeWise
Demand Forecasting Model	Predicts future product demand by analyzing historical sales records, customer purchasing patterns, and production information.
Shelf-Life Prediction Model	Determines product freshness conditions by analyzing production dates, product type, and shelf-life information.
Inventory Analytics Model	Evaluates stock movement, product availability, and inventory patterns to identify possible concerns.
Waste Analytics Model	Analyzes waste records to determine frequently wasted products and possible causes of product loss.
Production Optimization Model	Provides recommended production quantities based on predicted demand and inventory conditions.
Recommendation Engine	Generates suggestions related to production adjustments, inventory

	actions, and waste reduction strategies.
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3.1.5 Data Analytics Tools

Data analytics allows BakeWise to transform raw bakery information into meaningful operational insights.

The system analyzes collected information such as sales records, production data, inventory movement, and waste information.

Through data analytics, BakeWise can identify:

Product demand trends, frequently wasted products, inventory conditions, and production performance.

The results are presented through dashboards and reports that allow managers to understand bakery operations and make informed decisions.

3.1.6 User Interface Design Tools

The user interface of BakeWise is designed using a user-centered approach to ensure that bakery personnel can easily understand and operate the system.

Since users have different responsibilities, the interface is designed according to their specific roles.

Sales personnel require simple sales recording functions, inventory staff

require stock monitoring features, production staff require production tracking tools, and managers require analytical dashboards.

The system follows a responsive design approach to ensure compatibility with desktop computers, laptops, tablets, and mobile devices.

Table 4. User Interface Components of BakeWise

Interface Component	Purpose in BakeWise
Login Interface	Provides secure user authentication and identifies user access levels.
Dashboard Interface	Displays sales performance, inventory conditions, waste trends, and AI recommendations.
Sales Management Interface	Allows recording and monitoring of product sales transactions.
Inventory Management Interface	Allows monitoring of product availability and stock conditions.
Production Management Interface	Allows recording of production activities and viewing production recommendations.

AI Forecasting Interface	Displays predicted demand and AI-generated production suggestions.
Shelf-Life Monitoring Interface	Displays product freshness status and expiration alerts.
Waste Analytics Interface	Presents waste patterns and possible reduction strategies.
Reporting Interface	Generates reports for managers and future corporate monitoring.

3.2 Technical Description of BakeWise

3.2.1 Product Description

BakeWise is an Artificial Intelligence-powered bakery management system developed to improve bakery operations through intelligent data analysis and decision support.

The system addresses operational challenges related to bread waste, demand estimation, inventory monitoring, and production planning.

Unlike conventional systems that only record business transactions, BakeWise analyzes collected operational information and provides recommendations that help managers improve decisions.

The system consists of interconnected modules that manage sales, inventory, production, waste, forecasting, shelf-life monitoring, and reporting.

The initial implementation focuses on Rose Bakeshop – Lasang Branch, Davao City. The system is designed with scalability features to support future integration into other Rose Bakeshop branches after approval from Jathnier Corporation.

3.2.2 Economic Use of the Product

BakeWise provides economic value by helping bakery businesses reduce operational losses and improve resource utilization.

The demand forecasting feature reduces excessive production by helping managers estimate the quantity of products needed based on historical patterns.

The inventory analytics feature minimizes losses by identifying products that require attention and improving stock monitoring.

The waste analytics and product repurposing features allow businesses to maximize the value of unsold products and reduce unnecessary disposal.

Through improved planning and monitoring, BakeWise can contribute to better operational efficiency and sustainable bakery practices.

3.2.3 User Requirements

BakeWise is designed according to the needs of different bakery personnel.

The Branch Manager requires access to dashboards, reports, and AI recommendations to support operational decisions.

Production staff require tools for recording production activities and viewing recommended quantities.

Inventory staff require features for monitoring stock availability, product movement, and shelf-life conditions.

Sales staff require an efficient interface for recording transactions.

Administrators require tools for user management and system maintenance.

Future Jathnier Corporation users require centralized monitoring capabilities to review branch performance and operational trends.

3.3 System Requirements

3.3.1 Hardware Requirements

BakeWise requires hardware capable of running a web-based application.

The system can operate using:

Desktop computers, laptops, tablets, and mobile devices.

Recommended hardware includes devices with sufficient processing

capability, adequate memory, and stable internet connectivity.

For future multi-branch integration, additional server resources may be required depending on the number of connected branches.

3.3.2 Software Requirements

BakeWise requires software components that support system operation and maintenance.

These include:

Web browsers for accessing the system, database management systems for storing information, server environments for application processing, and Artificial Intelligence libraries for predictive analysis.

The combination of these technologies allows BakeWise to function as an integrated bakery management platform.

3.4 System Features

3.4.1 User Management Module

The User Management Module manages system accounts, authentication, and user permissions.

It ensures that each user can only access functions related to their responsibilities.

3.4.2 Sales Management Module

The Sales Management Module records daily sales transactions and maintains historical sales information.

The data collected by this module serves as input for demand forecasting and sales analysis.

3.4.3 Inventory Management Module

The Inventory Management Module monitors available products, stock movement, and product conditions.

It helps identify inventory concerns and supports better stock management.

3.4.4 Production Management Module

The Production Management Module records production activities and provides recommendations based on demand forecasts and inventory information.

3.4.5 Demand Forecasting Module

The Demand Forecasting Module uses Artificial Intelligence to analyze historical sales data and predict future product requirements.

This feature supports production planning and reduces overproduction.

3.4.6 Shelf-Life Prediction Module

The Shelf-Life Prediction Module monitors product freshness and identifies products approaching expiration.

This helps prevent unnecessary waste.

3.4.7 Waste Analytics Module

The Waste Analytics Module analyzes waste records to identify waste patterns and possible improvement opportunities.

3.4.8 Product Repurposing Recommendation Module

This module provides suggestions for maximizing safe unsold products through alternative uses.

Examples include bread crumbs, croutons, and bread pudding.

3.4.9 Dashboard and Reporting Module

The Dashboard and Reporting Module presents summarized operational information through charts, reports, and analytical displays.

It supports management monitoring and decision-making.

3.5 System Architecture

The BakeWise system architecture follows a layered structure composed of the user layer, application layer, database layer, Artificial Intelligence layer, and management layer.

3.5.1 AI Processing Architecture of BakeWise

The AI architecture begins with data collection from sales, production, inventory, waste records, and product information.

The collected data are stored in a centralized database where they are processed by the AI engine.

The AI engine performs demand forecasting, shelf-life prediction, inventory analytics, production optimization, and waste analysis.

The processed results are presented through reports, dashboards, alerts, and recommendations for management decision-making.

3.5.2 Main Office and Jathnier Corporation Integration

BakeWise is designed for future expansion beyond the Lasang Branch.

After successful implementation and approval of Jathnier Corporation, the system may integrate with other Rose Bakeshop branches within Davao City.

The future architecture allows centralized monitoring of branch operations, performance comparison, and company-wide reporting.

3.5.3 Network Design

BakeWise uses a client-server architecture.

Users access the system through computers, tablets, or mobile devices.

The application server manages system operations, while the database server stores operational information.

The AI processing server analyzes collected data and generates recommendations.

The management dashboard provides summarized information for decision-making.

3.5.4 Network Security

Security measures are implemented to protect bakery information and user accounts.

The system applies authentication, role-based access control, password protection, database security, and backup procedures.

These security measures ensure that operational information remains confidential, accurate, and accessible only to authorized users.

CHAPTER IV: METHODOLOGY AND DESIGN

This chapter presents the methodology and system design approach used in the development of **BakeWise: Bread Waste Reduction and AI-Driven Knowledge for Efficient Waste Management, Inventory Analytics, Shelf-Life Prediction, and Production Enhancement**.

The chapter discusses the research design, data gathering procedures,

sampling technique, software development methodology, requirements analysis, system design, development process, system testing and validation, deployment and monitoring plan, and ethical considerations.

The methodology was designed to ensure that BakeWise is developed based on the actual operational needs of **Rose Bakeshop – Lasang Rose Branch, Davao City**. The study focuses on addressing bakery operational challenges, particularly in production planning, inventory management, shelf-life monitoring, demand forecasting, and reduction of bread waste.

The initial data gathering activities were conducted at Rose Bakeshop – Lasang Rose Branch to identify existing processes, operational difficulties, and system requirements. The gathered information served as the foundation for designing the proposed Artificial Intelligence-powered bakery management system.

The proposed BakeWise system will initially focus on the Lasang Rose Branch as the intended implementation site. The system will undergo development, testing, and validation based on the identified requirements. After successful implementation, evaluation, and approval by **Jathnier Corporation**, future integration may be considered for other Rose Bakeshop branches within Davao City to support

centralized monitoring and standardized bakery operations.

4.1 Research Design

The study utilizes developmental research, descriptive research, and mixed-method research approaches. These approaches were selected because the study involves the development of a software-based solution while requiring an understanding of existing bakery processes and user requirements.

The combination of these research designs allows the researchers to identify operational problems, gather necessary information, develop the proposed system, and evaluate its effectiveness.

4.1.1 Developmental Research

Developmental research was applied because the main objective of the study is to design and develop BakeWise as an Artificial Intelligence-powered bakery management system.

According to Richey and Klein (2007), developmental research focuses on the systematic creation, design, development, and evaluation of products, tools, or processes. This approach is appropriate for technology-based studies because it allows researchers to transform identified problems into practical solutions.

In this study, developmental research guided the creation of BakeWise from the initial identification of bakery challenges up to the development of the proposed system.

The developmental process includes requirements analysis, system design, database creation, Artificial Intelligence integration, software development, testing, and evaluation.

Through this approach, BakeWise is designed to provide solutions for common bakery concerns such as inaccurate demand estimation, excessive production, inventory monitoring difficulties, and product waste management.

4.1.2 Descriptive Research

Descriptive research was used to understand and document the current operational practices of Rose Bakeshop – Lasang Rose Branch.

This approach allowed the researchers to examine existing bakery procedures, including sales recording, production planning, inventory monitoring, product handling, and waste management.

The descriptive approach helped identify limitations in current practices. These include dependence on manual estimation, difficulty in predicting customer demand, challenges in monitoring product freshness, and limited data analysis capabilities.

The information obtained through descriptive research served as the basis for identifying the required features and functions of BakeWise.

4.1.3 Mixed Method Research

The study applied a mixed-method research approach by combining qualitative and quantitative methods.

The qualitative component involved interviews, consultations, and observations conducted during the initial data gathering phase. These activities helped the researchers understand the experiences, challenges, and expectations of bakery personnel.

The quantitative component will be applied during system evaluation through software testing, user assessment, and system performance measurement.

The combination of qualitative and quantitative approaches provides a complete understanding of bakery operations and ensures that BakeWise is developed according to actual operational requirements.

4.2 Data Gathering Procedures

The initial data gathering procedure was conducted to obtain the necessary information required for the planning and development of BakeWise.

The data gathering was conducted at **Rose Bakeshop – Lasang Rose Branch, Davao City**, which serves as

the official target implementation site of the proposed system.

The purpose of the initial data gathering was to understand existing bakery operations, identify operational problems, and determine the system requirements needed for developing an Artificial Intelligence-powered bakery management system.

The researchers conducted interviews and consultations with selected personnel of Rose Bakeshop – Lasang Rose Branch who were directly involved in daily operations.

The participants provided information regarding current procedures in:

- Sales monitoring
- Inventory management
- Production planning
- Product monitoring
- Waste handling

The gathered responses helped identify challenges related to demand estimation, inventory monitoring, shelf-life management, and reducing unsold bakery products.

The researchers also conducted observation of the existing bakery workflow to understand how employees perform daily activities.

The observation focused on:

Product preparation, inventory checking, sales recording, production scheduling, and handling of unsold products.

Through observation, the researchers identified areas where Artificial Intelligence and automated analysis could provide operational support.

Available operational information and existing records were also considered during the gathering process. These included product information, sales-related information, inventory procedures, production practices, and waste management processes.

The collected information was analyzed and converted into system requirements. These requirements became the foundation for designing BakeWise features such as demand forecasting, shelf-life prediction, inventory analytics, production recommendation, waste analysis, and reporting.

The initial data gathering does not represent the final implementation, testing, or evaluation of the system. The gathered information was only used as the basis for designing and developing the proposed BakeWise system.

4.3 Sampling Technique

The study utilized purposive sampling in selecting participants who contributed information during the requirements analysis stage.

Purposive sampling was selected because the researchers required participants with direct knowledge and experience regarding bakery operations.

According to Palinkas et al. (2015), purposive sampling allows researchers to intentionally select individuals who possess relevant knowledge related to the objectives of the study.

The participants were selected based on their responsibilities within Rose Bakeshop – Lasang Rose Branch.

The selected participants include:

Participant	Contribution
Branch Manager	Provided information regarding operational management, decision-making processes, and system expectations.
Production Personnel	Provided information regarding product preparation, production scheduling, and quantity estimation.
Inventory Personnel	Provided information regarding stock monitoring, product movement, and shelf-life concerns.
Sales Personnel	Provided information

	regarding sales recording and customer demand patterns.
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The information gathered from these participants contributed to identifying the required functions and features of BakeWise.

Their operational knowledge ensured that the proposed system design was aligned with the actual needs of bakery operations.

4.4 Software Development Methodology

The development of BakeWise follows the **Agile Scrum methodology** as the primary software development approach.

Agile Scrum was selected because it supports iterative development, continuous improvement, and flexibility in responding to changing system requirements.

The integration of Artificial Intelligence features requires continuous evaluation and refinement because forecasting accuracy, recommendations, and analytical outputs depend on the quality of collected data and system performance.

According to Schwaber and Sutherland (2020), Scrum allows development teams to create complex products

through incremental development cycles called sprints.

Each sprint focuses on planning, developing, testing, reviewing, and improving specific system components.

The Agile Scrum methodology allows BakeWise to be developed systematically while maintaining continuous improvement based on feedback and evaluation.

4.4.1 Agile Scrum Development Process

The Agile Scrum development process of BakeWise consists of the following stages:

Planning

The planning stage identifies project objectives, system scope, development priorities, and expected outputs.

During this stage, the researchers analyze the operational concerns identified from Rose Bakeshop – Lasang Rose Branch and determine the system features required to address these concerns.

The planned system modules include:

1. User Management Module
2. Sales Management Module
3. Inventory Management Module
4. Production Management Module
5. Demand Forecasting Module
6. Shelf-Life Prediction Module
7. Waste Analytics Module

8. Recommendation Module
9. Reporting Dashboard

Requirements Analysis

The requirements analysis stage determines the functional and non-functional requirements of BakeWise.

The researchers analyze the gathered information to identify what the system must accomplish and what qualities it must maintain.

The requirements identified during this stage guide the system design and development process.

System Design

The system design stage focuses on creating the structure and components of BakeWise.

The researchers design:

1. System architecture
2. Database structure
3. User interface
4. Artificial Intelligence workflow
5. Data flow process
6. System diagrams

This stage ensures that all system components are properly organized before development.

Development

The development stage involves creating the actual BakeWise system.

The researchers develop the user interface, database, system modules, and Artificial Intelligence components.

The system is developed in smaller increments to allow continuous testing and improvement.

Testing

The testing stage evaluates whether BakeWise functions according to the identified requirements.

Testing focuses on:

1. Functionality
2. Usability
3. Reliability
4. Performance
5. Security

The results of testing are used to identify areas requiring improvement.

Deployment

The deployment stage involves preparing BakeWise for implementation at Rose Bakeshop – Lasang Rose Branch.

This includes system configuration, database preparation, user setup, and orientation.

Continuous Improvement

Continuous improvement allows future enhancement of BakeWise based on user feedback and operational requirements.

After successful development, testing, validation, and approval of Jathnier Corporation, future integration with other Rose Bakeshop branches may be considered.

Agile Scrum Development Model of BakeWise

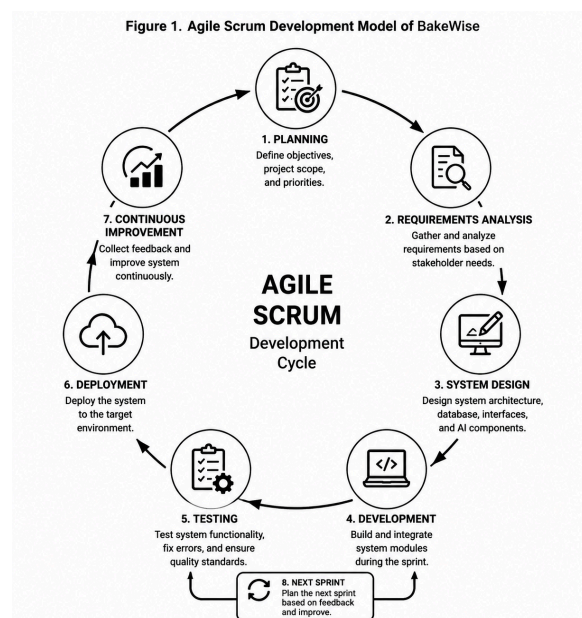


Figure 1. Agile Scrum Development Model of BakeWise

The figure illustrates the iterative development cycle used in creating BakeWise. The process allows continuous improvement by using feedback from previous development cycles to improve future system versions.

4.5 Requirements Analysis

Requirements analysis was conducted to identify the necessary functions, features, and quality attributes required for the development of BakeWise. The requirements were determined based on the initial data gathering conducted at **Rose Bakeshop – Lasang Rose Branch, Davao City**.

The purpose of requirements analysis is to ensure that the proposed system addresses the actual operational needs of the bakery. The identified requirements serve as the foundation for designing the system architecture, database structure, Artificial Intelligence components, and user interface.

The requirements are divided into two categories: functional requirements and non-functional requirements. Functional requirements describe the services and operations that BakeWise must perform, while non-functional requirements define the quality standards that ensure system reliability, security, usability, and scalability.

4.5.1 Functional Requirements

Functional requirements describe the specific features and activities that BakeWise must perform. These requirements define how users interact with the system and how the system processes bakery information.

The functional requirements were identified based on the operational processes of Rose Bakeshop – Lasang Rose Branch, particularly in sales

monitoring, inventory management, production planning, waste monitoring, and management decision-making.

Table 5. Functional Requirements of BakeWise

Functional Requirement	Description
User Authentication	The system shall provide a secure login process that allows authorized users to access BakeWise using assigned accounts and credentials.
User Management	The system shall allow administrators to create, manage, update, and control user access according to assigned roles and responsibilities.
Sales Management	The system shall allow users to record sales transactions and maintain historical sales information that will serve as input for AI analysis.
Inventory Management	The system shall monitor available products, inventory movement, stock conditions, and

	product availability status.
Production Management	The system shall record production activities and provide information that supports production planning and monitoring.
Demand Forecasting	The system shall analyze historical sales data and generate predictions regarding future product demand.
Shelf-Life Prediction	The system shall monitor product freshness by analyzing product information, production dates, and shelf-life data.
Waste Monitoring	The system shall record and analyze wasted products, quantities, and reasons for disposal.
Waste Analytics	The system shall identify waste patterns and provide insights regarding products that frequently contribute to waste.

Production Recommendation	The system shall provide recommendations regarding appropriate production quantities based on demand forecasting and inventory conditions.
Product Repurposing Recommendation	The system shall suggest possible alternative uses for safe unsold bakery products to reduce unnecessary disposal.
Dashboard Monitoring	The system shall display summarized information regarding sales performance, inventory status, waste trends, and AI-generated insights.
Report Generation	The system shall generate reports that assist managers in reviewing bakery performance and making informed decisions.

4.5.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance expectations of BakeWise. These requirements ensure that the system operates efficiently and provides a reliable experience for users.

Table 6. Non-Functional Requirements of BakeWise

Non-Functional Requirement	Description
Usability	The system shall provide a simple and understandable interface that allows bakery personnel to operate the system with minimal difficulty.
Reliability	The system shall maintain accurate processing and storage of bakery information to ensure dependable results.
Security	The system shall protect business information and user accounts through authentication, access control, and secure data handling.
Performance Efficiency	The system shall process user requests, data analysis, and report generation within an

	acceptable response time.
Maintainability	The system shall allow future updates, improvements, and technical maintenance.
Scalability	The system shall support future expansion and possible integration with other Rose Bakeshop branches after Jathnir Corporation approval.
Compatibility	The system shall be accessible through common web browsers and different devices such as computers, tablets, and mobile devices.
Availability	The system shall be accessible whenever authorized users require bakery information and operational reports.
Data Integrity	The system shall ensure that stored information

	remains accurate, consistent, and protected from unauthorized modification.
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4.6 System Design

The system design presents the overall structure, components, and workflow of BakeWise. It explains how bakery information enters the system, how Artificial Intelligence processes the collected data, and how the generated results are presented to users.

The design of BakeWise follows a structured architecture that connects operational data, database management, Artificial Intelligence processing, and user decision support.

The system design consists of the following components:

1. Conceptual Framework
2. Use Case Diagram
3. Context Flow Diagram
4. Data Flow Diagram
5. System Architecture Design

4.6.1 Conceptual Framework

The conceptual framework of BakeWise follows the Input-Process-Output-Benefit (IPO-B) model.

The IPO-B model explains how operational information from Rose

Bakeshop is transformed into useful recommendations through Artificial Intelligence processing.

The framework begins with the collection of bakery information, followed by system analysis and processing, which produces outputs that support management decisions.

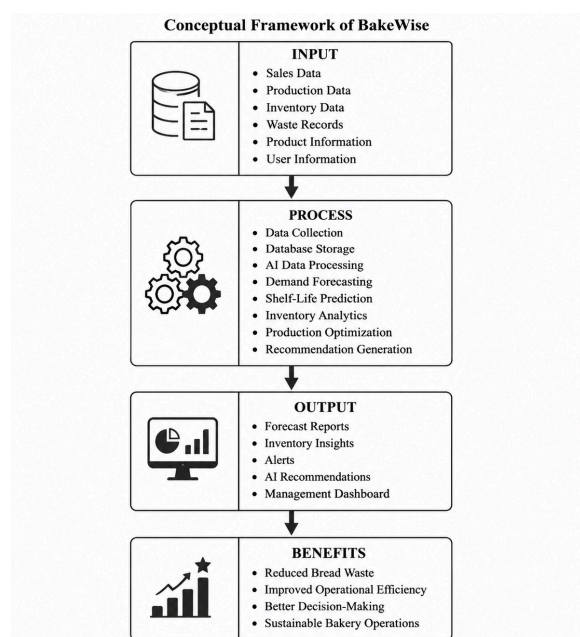


Figure 2. Conceptual Framework of BakeWise

The input component represents the information gathered from bakery operations. These data serve as the foundation for Artificial Intelligence analysis.

The process component represents the activities performed by BakeWise, including database management, data

analysis, forecasting, prediction, and recommendation generation.

The output component represents the information generated by the system, including reports, alerts, dashboards, and recommendations.

The benefit component represents the expected improvements from using BakeWise, including reduced product waste, improved efficiency, and better operational decision-making.

4.6.2 Use Case Diagram

The Use Case Diagram illustrates the interaction between users and the BakeWise system.

The system involves different users with specific responsibilities. Each user interacts with the system according to their operational role.

The identified actors include:

- Administrator
- Branch Manager
- Sales Personnel
- Inventory Personnel
- Production Personnel
- Future Jathnier Corporation Management

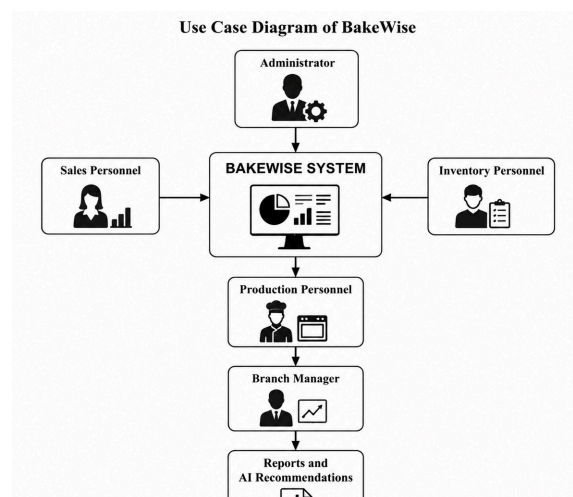


Figure 3. Use Case Diagram of BakeWise

The diagram shows how different users interact with BakeWise. Sales personnel provide transaction data, inventory personnel manage product information, production personnel record production activities, and managers review reports and recommendations.

The administrator manages user access and system settings. Future Jathnier Corporation management may access centralized reports after successful system integration.

4.6.3 Context Flow Diagram

The Context Flow Diagram presents the relationship between BakeWise and external entities.

It shows how information is received from users and how processed information is returned through system outputs.

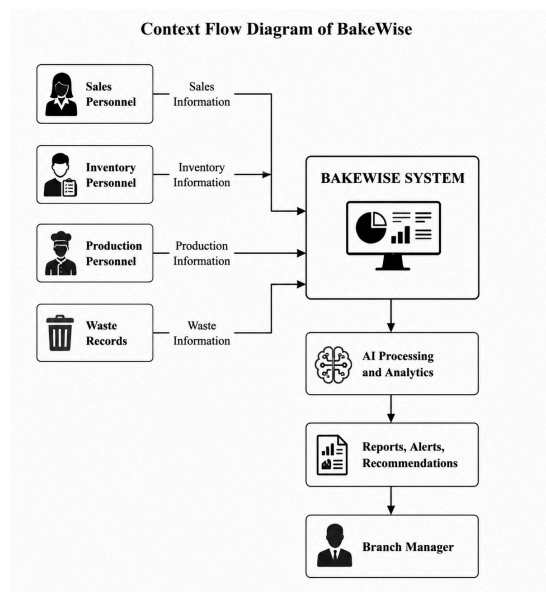


Figure 4. Context Flow Diagram of BakeWise

The diagram demonstrates that BakeWise receives operational information from different bakery personnel. The system processes the information through database management and Artificial Intelligence analysis before generating useful outputs for management decision-making.

4.6.4 Data Flow Diagram

The Data Flow Diagram explains how information moves inside BakeWise.

The system processes data through different modules, including sales management, inventory management, production management, waste analytics, and Artificial Intelligence processing.

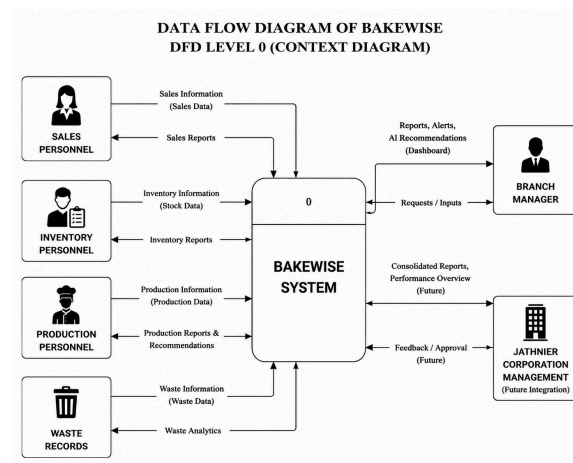


Figure 4.6.1 – DFD Level 0 of BakeWise

The DFD Level 0 presents the overall data flow and interaction between the external entities and the BakeWise system. It illustrates how operational information from different bakery personnel is collected, processed, and transformed into meaningful outputs for decision-making.

The Sales Personnel provide sales information, including transaction records and product sales data, which serve as an input for analyzing customer demand patterns. The Inventory Personnel submit inventory-related information such as available stocks, product movement, and inventory status. The Production Personnel provide production information, including prepared products and production records, while Waste Records provide information regarding discarded products and waste quantities.

The BakeWise System serves as the central processing unit that receives, stores, and analyzes all collected operational data. Through Artificial Intelligence processing and analytics, the system generates outputs such as reports, alerts, inventory insights, and AI-based recommendations.

The Branch Manager receives the generated dashboards, reports, and recommendations to support operational decision-making. The future integration with Jathnir Corporation Management allows centralized monitoring and evaluation of branch performance after

successful implementation, validation, and approval of the system.

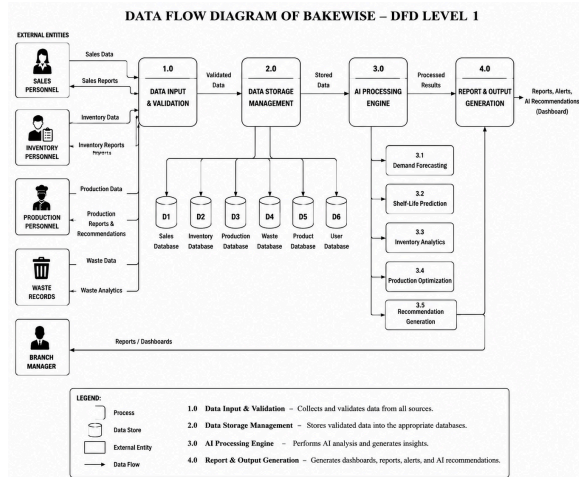


Figure 4.6.2 – DFD Level 1 of BakeWise

The DFD Level 1 provides a detailed representation of the internal processes and data flow within the BakeWise system. It expands the major functions of the system into specific processes, including data input and validation, data storage management, Artificial Intelligence processing, and report generation.

The Data Input and Validation process collects information from different sources, including sales data, inventory data, production records, and waste information. The collected data are verified and prepared to ensure accuracy before being stored in the system database.

The Data Storage Management process organizes and maintains collected information through different databases,

including the Sales Database, Inventory Database, Production Database, Waste Database, Product Database, and User Database. These databases serve as the foundation for Artificial Intelligence analysis and system operations.

The AI Processing Engine analyzes the stored information through different intelligent modules. The Demand Forecasting module identifies future product demand based on historical sales patterns. The Shelf-Life Prediction module evaluates product freshness and identifies products approaching expiration. The Inventory Analytics module examines stock movement and inventory conditions. The Production Optimization module provides recommendations regarding appropriate production quantities. The Recommendation Generation module produces AI-based suggestions to support bakery operations.

The Report and Output Generation process converts the analyzed information into understandable outputs, including dashboards, reports, alerts, and recommendations. These outputs are provided to the Branch Manager to support informed decision-making and improve bakery efficiency.

The DFD Level 1 demonstrates how BakeWise transforms raw operational data into intelligent business insights through database management and Artificial Intelligence processing. It supports the goal of reducing bread waste, improving inventory control,

optimizing production activities, and preparing the system for future integration across Rose Bakeshop branches under Jathnier Corporation.

4.6.5 System Architecture Design

The system architecture design presents the overall structure and relationship of the different components that comprise the BakeWise system. It describes how users interact with the system, how operational information is processed, how data are stored, how Artificial Intelligence analyzes information, and how results are presented to support bakery decision-making.

BakeWise follows a layered system architecture to ensure organized communication between users, system processes, databases, Artificial Intelligence components, and management outputs. The architecture is designed to support efficient data processing, secure information management, and future scalability for possible integration with other Rose Bakeshop branches under Jathnier Corporation.

The first layer of the architecture is the User Layer, which represents the individuals who interact with the BakeWise system. This layer includes Sales Personnel, Inventory Personnel, Production Personnel, Branch Manager, and future Jathnier Corporation management users. Users provide operational information such as sales

transactions, inventory updates, production records, and waste information. They also access system-generated reports and recommendations depending on their assigned responsibilities.

The second layer is the Application Layer, which manages the main operations of BakeWise. This layer handles user authentication, data processing, system functions, dashboard presentation, and communication between users and the system database. It serves as the connection between the user interface and the internal processes of the system.

The third layer is the Database Layer, which stores and manages all information collected from bakery operations. The database contains sales records, inventory information, production records, waste records, product details, and user information. This layer ensures that operational data are properly organized, secured, and available for future analysis.

The fourth layer is the Artificial Intelligence Layer, which serves as the analytical component of BakeWise. This layer processes collected information through different AI-based functions, including demand forecasting, shelf-life prediction, inventory analytics, production optimization, and recommendation generation. Through these processes, the system transforms raw bakery information into meaningful

insights that can assist managers in making operational decisions.

The final layer is the Management Layer, which presents the processed information through dashboards, reports, alerts, and AI-generated recommendations. This layer allows the Branch Manager to monitor bakery performance, evaluate operational conditions, and make informed decisions. In the future, after successful implementation and approval from Jathnier Corporation, this layer may support centralized monitoring of multiple Rose Bakeshop branches.

Figure 4.6.3

System Architecture Design of BakeWise

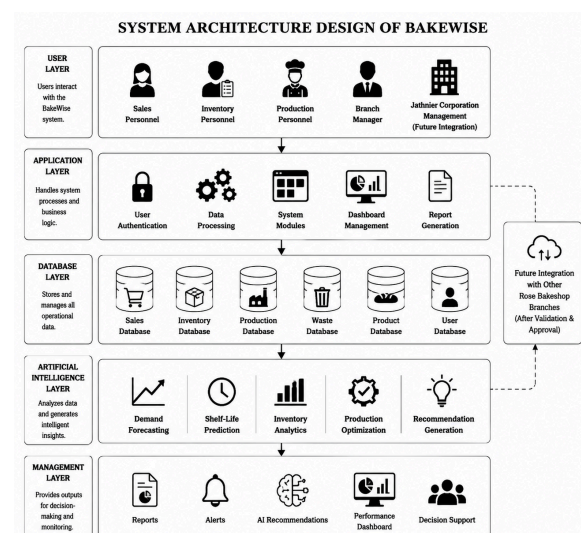


Figure 4.6.3. System Architecture Design of BakeWise

The figure illustrates the complete structure of BakeWise and shows how information moves from users to system processing and finally to management decision support. The architecture ensures that each component performs a specific function while maintaining communication between different layers. This design allows BakeWise to operate as an intelligent bakery management system capable of improving production planning, inventory control, and waste reduction.

4.7 Development

The development phase focuses on transforming the approved system requirements and design specifications into a functional BakeWise system. This phase follows the Agile Scrum methodology, where system development is performed through continuous improvement cycles. Each development cycle focuses on designing, developing, evaluating, and improving specific system components based on the identified operational requirements of Rose Bakeshop – Lasang Rose Branch.

The development process begins with the creation of the system interface. The user interface of BakeWise is designed to provide accessibility, simplicity, and ease of use for bakery personnel with different operational responsibilities. Since the system involves multiple users, including sales personnel, inventory personnel, production

personnel, and branch managers, the interface is structured according to their specific roles and system requirements.

The database development focuses on organizing and managing the information required by BakeWise. The database is designed to store essential bakery information, including sales records, inventory information, production activities, waste records, product details, and user account information. Proper database organization ensures that operational information can be efficiently retrieved, analyzed, and processed for reporting and Artificial Intelligence functions.

The Artificial Intelligence component serves as the main analytical feature of BakeWise. The system processes historical and current bakery information to generate useful insights that support operational decision-making. Demand forecasting analyzes previous sales patterns to estimate future product requirements. Shelf-life prediction evaluates product information and freshness conditions to identify products that require attention. Inventory analytics examines stock movement and availability status, while production optimization provides recommendations regarding appropriate production quantities. The recommendation engine generates suggestions that support waste reduction and improve bakery efficiency.

The development process also includes the integration of reporting and

dashboard features. These features allow managers to view summarized information regarding sales performance, inventory conditions, production activities, waste patterns, and Artificial Intelligence-generated recommendations.

Through the Agile Scrum development process, each completed component is reviewed, tested, and improved before proceeding to the next development cycle. This approach ensures that the developed BakeWise system remains aligned with the operational requirements of Rose Bakeshop – Lasang Rose Branch.

4.7.1 Prototype Description of BakeWise

The BakeWise prototype represents the initial visual and functional design model of the proposed Artificial Intelligence-powered bakery management system. The prototype was developed based on the system requirements identified during the initial data gathering conducted at Rose Bakeshop – Lasang Rose Branch, Davao City.

The prototype serves as a representation of how users will interact with BakeWise and how the different system modules will support bakery operations. It demonstrates the planned features, user interface structure, system navigation, and information presentation of the proposed platform.

The prototype emphasizes usability, accessibility, and mobile responsiveness to ensure that bakery personnel and management can easily monitor operations and access important information. The design follows the identified functional requirements of BakeWise, including sales monitoring, inventory management, production monitoring, waste analysis, Artificial Intelligence analytics, and reporting.

The system prototype applies role-based access to ensure that each user can access features appropriate to their responsibilities. Sales personnel can manage sales-related information, inventory personnel can monitor product availability, production personnel can record production activities, and the branch manager can access reports, dashboards, and Artificial Intelligence-generated recommendations.

The prototype consists of major system interfaces, including the Login Interface, Dashboard Interface, Sales Management Interface, Inventory Management Interface, Production Management Interface, Waste Monitoring Interface, Artificial Intelligence Analytics Interface, and Reporting Dashboard Interface.

4.7.2 Login Interface Prototype

The Login Interface serves as the entry point of the BakeWise system. It provides security and access control by requiring authorized users to enter their

assigned credentials before accessing system functions.

The purpose of this interface is to ensure that only authorized personnel can access bakery operational information. User authentication also allows the system to identify the user's role and provide appropriate permissions based on assigned responsibilities.

The interface contains username and password fields, login verification, and account access controls. This supports the security requirements of BakeWise by protecting confidential bakery information from unauthorized access.

Figure 4.7.1

BakeWise Login Interface Prototype

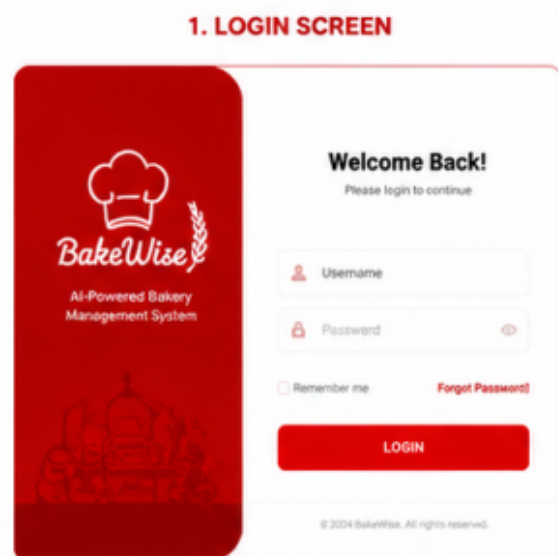


Figure 4.7.1. BakeWise Login Interface Prototype

4.7.3 Dashboard Interface Prototype

The Dashboard Interface serves as the main monitoring interface of BakeWise. It provides users, particularly the Branch Manager, with a summarized overview of bakery operations.

The dashboard displays important information such as sales performance, inventory status, production information, waste trends, and Artificial Intelligence-generated insights. These visual summaries allow managers to quickly understand the current operational condition of the bakery.

Through the dashboard, management can monitor performance indicators and use generated information as support for operational planning and decision-making.

Figure 4.7.2

BakeWise Dashboard Interface Prototype



Figure 4.7.2. BakeWise Dashboard Interface Prototype

4.7.4 Sales Management Interface Prototype

The Sales Management Interface allows authorized personnel to record and manage sales-related information.

This module collects transaction data that serves as an important input for Artificial Intelligence analysis. Historical sales records allow the system to identify demand patterns and generate future product demand forecasts.

The interface allows users to record product information, quantities sold, transaction details, and sales dates. The collected information is stored in the database and used for reporting and demand analysis.

Figure 4.7.3

BakeWise Sales Management Interface Prototype

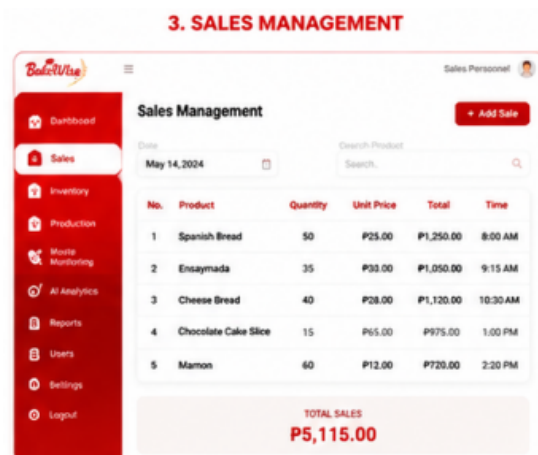


Figure 4.7.3. BakeWise Sales Management Interface Prototype

4.7.5 Inventory Management Interface Prototype

The Inventory Management Interface allows users to monitor available products, stock levels, and inventory conditions.

This module supports inventory control by providing updated information regarding product availability and movement. It helps reduce shortages and excessive inventory accumulation by allowing personnel to monitor current stock conditions.

The collected inventory information is also used by Artificial Intelligence features to analyze stock patterns and support shelf-life monitoring.

Figure 4.7.4

BakeWise Inventory Management Interface Prototype

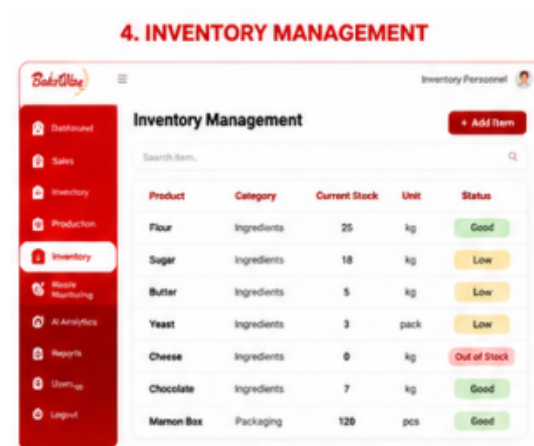


Figure 4.7.4. BakeWise Inventory Management Interface Prototype

4.7.6 Production Management Interface Prototype

The Production Management Interface allows production personnel to record and monitor bakery production activities.

This module records production schedules, planned quantities, actual production quantities, and production status. The information collected supports Artificial Intelligence analysis for production optimization.

Through this interface, BakeWise assists bakery personnel in determining appropriate production levels based on demand forecasts and inventory conditions.

Figure 4.7.5

BakeWise Production Management Interface Prototype

Product	Planned Qty	Produced Qty	Time Started	Status
Spanish Bread	200	200	6:00 AM	Completed
Ensaymada	150	150	6:30 AM	Completed
Cheese Bread	150	140	7:00 AM	In Progress
Chocolate Cake	10	10	8:00 AM	Completed
Mamon	300	280	9:00 AM	In Progress

Figure 4.7.5. BakeWise Production Management Interface Prototype

4.7.7 Waste Monitoring Interface Prototype

The Waste Monitoring Interface allows users to record and analyze bakery waste information.

This module collects information regarding discarded products, waste quantity, product category, and possible causes of waste. The collected information allows the system to identify waste patterns and provide recommendations that support waste reduction.

The module supports the main objective of BakeWise by helping reduce unnecessary product disposal and improve resource utilization.

Figure 4.7.6

BakeWise Waste Monitoring Interface Prototype

Product	Quantity (kg)	Reason	Time
Spanish Bread	5.0	Expired	11:00 AM
Ensaymada	3.2	Overproduction	1:00 PM
Cheese Bread	2.8	Damaged	1:30 PM
Mamon	2.5	Overproduction	2:00 PM
Chocolate Cake	5.0	Expired	3:00 AM

TOTAL WASTE TODAY: 18.5 kg

Figure 4.7.6. BakeWise Waste Monitoring Interface Prototype

4.7.8 Artificial Intelligence Analytics Interface Prototype

The Artificial Intelligence Analytics Interface represents the intelligent processing component of BakeWise.

This interface presents AI-generated outputs, including demand forecasts, shelf-life predictions, inventory insights, production recommendations, and waste reduction suggestions.

The purpose of this module is to transform collected bakery information into meaningful insights that support operational planning and management decisions.

The AI analytics feature provides recommendations while maintaining human decision-making responsibility. Final operational decisions remain under the authority of authorized bakery personnel.

Figure 4.7.7

BakeWise Artificial Intelligence Analytics Interface Prototype



Figure 4.7.7. BakeWise Artificial Intelligence Analytics Interface Prototype

4.7.9 Reporting and Management Dashboard Prototype

The Reporting and Management Dashboard provides summarized reports generated from bakery operational data and Artificial Intelligence analysis.

This interface allows management to evaluate sales performance, inventory conditions, production activities, waste trends, and system recommendations.

The generated reports support operational evaluation and strategic decision-making. For future expansion, this module may serve as the foundation for centralized monitoring by Jathnir Corporation after successful system validation, approval, and implementation.

Figure 4.7.8

BakeWise Reporting and Management Dashboard Prototype



Figure 4.7.8. BakeWise Reporting and Management Dashboard Prototype

4.8 System Testing and Validation

System testing and validation are important stages in ensuring that BakeWise performs according to the identified requirements. The purpose of testing is to determine whether the developed system functions correctly, produces accurate outputs, and provides an acceptable user experience.

Testing will be conducted after the development of the system prototype. The researchers will evaluate different aspects of BakeWise, including functionality, usability, reliability, performance, and security.

Functional testing will determine whether each system feature operates properly based on its intended purpose. This includes testing user authentication, sales recording, inventory monitoring, production recording, Artificial Intelligence analysis, report generation, and dashboard display.

Usability testing will evaluate whether bakery personnel can easily understand and operate the system. Since BakeWise will be used by employees with different responsibilities, the system interface must provide clear navigation and understandable functions.

Performance testing will examine the ability of BakeWise to process information efficiently. This includes evaluating system response time, data processing capability, and report generation performance.

Security testing will evaluate the protection of user accounts and business information. The researchers will ensure that access to system functions is controlled through authentication and assigned user permissions.

Validation will involve gathering feedback from selected users of Rose Bakeshop – Lasang Rose Branch. Their evaluation will determine whether the system features address identified operational concerns and whether the system is suitable for bakery management operations.

4.8.1 Testing Strategy

The testing strategy of BakeWise follows a systematic approach to verify that the developed system satisfies both functional and quality requirements.

The researchers will conduct module testing to verify the performance of individual system components. Each module, including sales management, inventory management, production management, waste analytics, and Artificial Intelligence functions, will be evaluated separately before being integrated into the complete system.

Integration testing will be conducted to ensure that all system components communicate properly with each other. This process verifies that data entered from different modules can be processed correctly and used by Artificial Intelligence functions.

System testing will evaluate the overall operation of BakeWise as a complete system. This stage determines whether the system can successfully perform its intended functions under actual operational conditions.

User acceptance testing will be conducted with selected bakery personnel to determine whether BakeWise meets user expectations. The feedback obtained from users will serve as a basis for further improvements before final implementation.

4.9 Deployment and Monitoring Plan

The deployment and monitoring plan describes the procedures that will be followed after the completion of the development and validation of the BakeWise system. This phase focuses on preparing the system for actual implementation at **Rose Bakeshop – Lasang Rose Branch, Davao City** and ensuring that the system performs according to the identified operational requirements.

The deployment process will begin after the successful completion of system development and testing activities. The researchers will prepare the system environment, configure the necessary databases, establish user accounts, and ensure that all required system components are properly connected before implementation.

The initial deployment of BakeWise will be conducted only at **Rose Bakeshop –**

Lasang Rose Branch as the official implementation site. The Lasang Rose Branch will serve as the first operational environment where the proposed system will be evaluated based on actual bakery processes. Another Rose Bakeshop branch will not be used as a pilot system location.

Before deployment, the researchers will coordinate with authorized personnel of Rose Bakeshop to prepare the necessary operational information required by the system. Product information, user access details, and other required data will be organized to ensure that the system can properly perform its intended functions.

After system configuration, user orientation and training will be conducted to familiarize bakery personnel with the features and operation of BakeWise. The training process will focus on understanding the system interface, entering operational information, viewing reports, interpreting Artificial Intelligence recommendations, and utilizing generated insights for decision-making.

The monitoring phase will focus on observing system performance, user interaction, data accuracy, and overall effectiveness during operation. Feedback from bakery personnel will be collected to identify possible improvements in usability, functionality, and system reliability.

The researchers will continuously review system performance and address identified issues to improve the quality and effectiveness of BakeWise. Any necessary adjustments will be incorporated through the Agile Scrum continuous improvement process.

After successful implementation, testing, validation, and approval by **Jathnier Corporation**, the system may be considered for future integration into other Rose Bakeshop branches within Davao City. The expansion of BakeWise will allow centralized monitoring, standardized operational procedures, multi-branch reporting, and improved company-wide decision-making.

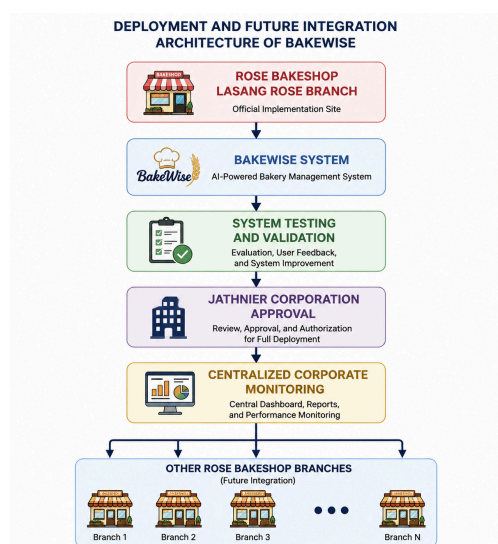


Figure 4.9.1. Deployment and Future Integration Architecture of BakeWise

The figure illustrates the planned deployment process of BakeWise. The system will initially operate at Rose Bakeshop – Lasang Rose Branch as the official implementation location. After

successful validation and approval from Jathnier Corporation, future integration may extend the system to other Rose Bakeshop branches.

This approach ensures that the system is properly evaluated before wider implementation. It allows the organization to verify system performance, improve operational procedures, and ensure that the system provides value before expansion.

4.10 Ethical Considerations

The researchers will ensure that all ethical principles are observed throughout the development of BakeWise: Bread Waste Reduction and AI-Driven Knowledge for Efficient Waste Management, Inventory Analytics, Shelf-Life Prediction, and Production Enhancement. Ethical considerations will guide the entire research process, including initial data gathering, system development, system testing, validation, and future implementation.

The study will prioritize the protection of participants, confidentiality of organizational information, responsible handling of collected data, and ethical use of Artificial Intelligence technology. The researchers will ensure that all activities are conducted with transparency, respect, and compliance with applicable laws and ethical research standards.

4.10.1 Organizational Permission and Research Authorization

Before conducting any research activity, the researchers will secure formal permission from the management of **Rose Bakeshop – Lasang Rose Branch, Davao City**. The researchers will provide a clear explanation regarding the purpose of the study, objectives of BakeWise, data requirements, expected procedures, and intended outcomes of the research.

The collection of operational information will only be conducted after receiving approval and authorization from the appropriate representatives of Rose Bakeshop. The researchers will ensure that the organization understands the purpose of the study and agrees to the use of necessary information for research and system development purposes.

The researchers will also acknowledge that the proposed system is initially intended only for the Lasang Rose Branch. Any future expansion, integration, or implementation of BakeWise in other Rose Bakeshop branches will only proceed after successful system development, testing, validation, and formal approval from **Jathnir Corporation**.

4.10.2 Informed Consent and Voluntary Participation

The researchers will obtain informed consent from all individuals who participate in interviews, consultations, observations, and future system evaluation activities.

Before participation, individuals will be provided with sufficient information regarding the purpose of the research, the nature of their involvement, the expected contribution, and the procedures involved in the study. Participants will be informed that their participation is voluntary and that they have the right to refuse participation or withdraw from the study at any time without experiencing any negative consequences.

The researchers will ensure that participants understand their role in the research before any information is collected. No participant will be forced or pressured to provide information beyond their willingness to participate.

4.10.3 Participant Rights and Protection

The researchers will respect and protect the rights of all participants involved in the study. Participants will be treated with fairness, respect, and dignity throughout the research process.

Participants will have the right to know how their information will be used, who will have access to collected information, and how their privacy will be protected. They will also have the right to request clarification regarding the research activities and their involvement.

The researchers will ensure that participation will not expose individuals to unnecessary risks, harm,

discrimination, or negative consequences.

4.10.4 Confidentiality of Information

The researchers will maintain strict confidentiality of all information collected during the study. Information obtained from Rose Bakeshop – Lasang Rose Branch, including sales information, inventory records, production procedures, waste records, product information, and business processes, will be treated as confidential.

Collected information will only be used for the purpose of developing and evaluating BakeWise. The researchers will not disclose business-related information to unauthorized individuals, organizations, or third parties.

Any research documents, collected responses, and system-related information will be stored securely and accessed only by authorized members of the research team.

4.10.5 Data Protection and Privacy Compliance

The researchers will comply with **Republic Act No. 10173, also known as the Data Privacy Act of 2012**, which protects the right to privacy and regulates the responsible processing of personal information.

The researchers will follow the principles of transparency, legitimate purpose, and proportionality in handling collected information. Only information necessary

for the development and evaluation of BakeWise will be collected.

Personal information obtained from participants will not be collected unnecessarily and will only be processed for the purpose explained during the informed consent process.

The researchers will apply appropriate data protection measures to prevent unauthorized access, loss, modification, or misuse of collected information. Digital files, research documents, and system-related records will be stored securely.

The developed BakeWise system will also consider data protection principles by implementing security features such as user authentication, controlled access permissions, and proper management of stored information.

4.10.6 Legal Basis and Compliance

The study will comply with applicable Philippine laws and ethical research standards related to information management and responsible technology development.

The primary legal basis of the study is **Republic Act No. 10173 or the Data Privacy Act of 2012**, which ensures responsible collection, processing, storage, and protection of information.

The study will also consider **Republic Act No. 10175 or the Cybercrime Prevention Act of 2012**, particularly in ensuring responsible handling of digital

information and preventing unauthorized access, alteration, or misuse of electronic data.

The researchers will also follow ethical research principles related to respect for participants, responsible data handling, and protection of stakeholder interests throughout the research process.

4.10.7 Responsible Artificial Intelligence Use

The researchers recognize the importance of responsible Artificial Intelligence implementation in the development of BakeWise.

The Artificial Intelligence components of BakeWise are designed to serve as decision-support tools that assist bakery personnel and management. The system will provide recommendations, forecasts, and analytical insights; however, final operational decisions will remain under the responsibility of authorized personnel.

The researchers will ensure that AI-generated outputs are properly reviewed and interpreted before being applied to actual bakery operations. The system will not independently make business decisions without human evaluation.

The researchers will also ensure that the information used for AI analysis is properly organized and reviewed to improve accuracy and reliability. The limitations of Artificial Intelligence will be acknowledged, and human judgment

will remain an essential part of operational decision-making.

4.10.8 Future System Expansion Control

Future expansion of BakeWise will only occur after successful system development, testing, validation, and approval from **Jathnier Corporation**.

The integration of BakeWise into other Rose Bakeshop branches will require proper evaluation of system performance, security, data management procedures, and operational compatibility.

Before implementing the system across additional branches, appropriate organizational approval, data protection considerations, and system readiness assessment will be conducted.

This ensures that future expansion will follow responsible technology adoption practices while maintaining confidentiality, security, and operational standards across all participating Rose Bakeshop branches.

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